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July 13, 2026

To: The Editorial Board

PLOS One

Dear Editors,

We are pleased to submit our manuscript entitled “WebDoctor: Personalized Doctor Outpatient Assistant via LoRA Routing and Merging in Internet Hospital Services” for consideration for publication in PLOS One. In this manuscript, we address a critical yet underexplored problem in clinical AI: the depersonalization of large language model (LLM) outputs in internet hospital settings. While recent advances in medical LLMs have shown impressive capabilities in clinical question answering and dialogue generation—as exemplified by Med-PaLM 2 and ClinicalCamel—existing systems produce generic responses that fail to capture individual physicians’ communication styles, departmental norms, and institutional constraints. This depersonalization fundamentally undermines physician trust and limits real-world clinical adoption, which is a gap that our work directly addresses.

Our proposed framework introduces a three-stage personalized adaptation pipeline: (i) doctor-specific Low-Rank Adaptation (LoRA) modules trained on individual physicians’ historical consultation records to encode personal response patterns; (ii) Doctor-Adaptive Routing (DAR), which dynamically selects the appropriate adapter at inference time through identity-based or query-based retrieval; and (iii) Sign-Consistent Fusion (SCF), an interference-aware merging technique that consolidates multiple adapters into a single deployment artifact while preserving individual performance. We constructed the Internet Hospital Doctor-Patient Dialogue (IHDPD) dataset spanning 121 clinical departments and 766 physicians from a real-world operational internet hospital. Evaluations across LLaMA-3.2-1B/3B and Qwen-2.5-7B demonstrate substantial improvements over strong baselines, with BLEU-4 scores rising from 3.2/5.1/8.7 to 6.9/10.4/14.2 and personalization ratings (1–5 scale) improving from 1.4/2.1/2.8 to 3.3/4.2/4.5 at 1B/3B/7B model scales, respectively. These results confirm that our framework enables trustworthy, physician-centered AI assistance that respects individual clinical identity.

We believe this manuscript is well-suited for PLOS One, which has a strong tradition of publishing interdisciplinary research at the intersection of digital health, clinical informatics, and artificial intelligence. Our work aligns directly with the journal’s scope in several ways. First, it addresses a practical challenge in internet hospital services. Second, the proposed framework is grounded in real-world clinical data and is designed for deployability within existing hospital information systems, bridging the gap between NLP research and clinical translation. Third, the personalization paradigm we introduce has broader implications for AI-assisted clinical decision support, medical education, and patient-provider communication—all central themes within the journal’s purview.

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal.

All authors have approved the manuscript and agree with its submission to PLOS One.

We appreciate your time and consideration and look forward to your response.

Sincerely,

Peng Zhao

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